EE1100: Basic Electrical Engineering
Tutorial 3: AC Circuits

Q1. Find the average and effective (rms) values of each wave-shape depicted in Fig. 1.

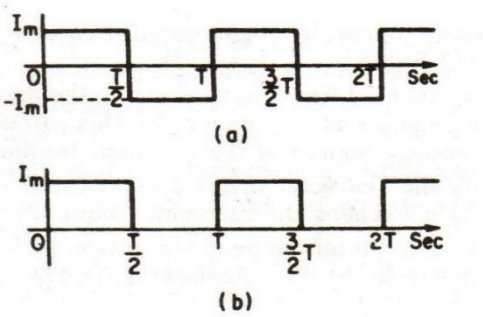


Fig. 1

[Ans.: (a) $I_{avg} = 0$, $I_{rms} = I_m$ (b) $I_{avg} = \frac{I_m}{2}$, $I_{rms} = \frac{I_m}{\sqrt{2}}$]

Q2. A voltage wave has the variation shown in Fig. 2. Find the average and the effective (rms) value of the voltage. If this voltage is applied to a 10 Ω resistance, find the dissipated power in Watts.

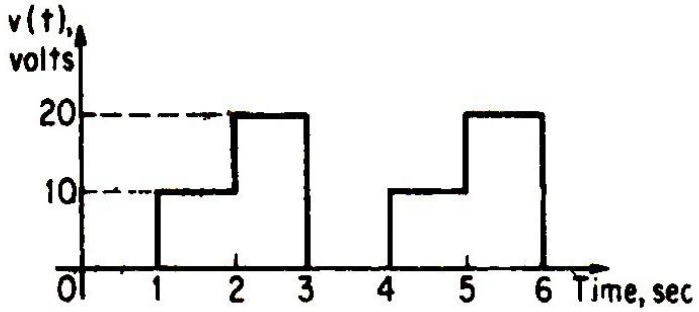


Fig. 2

[Ans.: $V_{avg} = 10\ V$, $V_{rms} = \frac{10\sqrt{15}}{3} = 12.909\ V$, $P = 16.667\ W$]

Q3. A circuit is composed of a resistance of 9 Ω and a series connected inductive reactance of 12 Ω. When a voltage *e(t)* is applied to the circuit, the resulting steady-state current is found to be

i(t) = 28.3 sin (377t).

(a) What is the value of complex impedance?
(b) Determine the time expression for the applied voltage.
(c) Find the value of the inductance in henrys.

[Ans.: (a) $Z = (9 + j12)$ Ω, (b) $v(t) = 424.5\sin(377t + 53.13°)$ V, (c) $L = 31.83\ mH$]

Q4. In an RC series circuit, to which is applied a voltage of 170 sin ωt, it is found that a steady-state current flows which leads the voltage by 30°. Find the effective voltage drops across the resistive and reactive elements.

[Ans.: $\bar{V}_R = 104.103\angle 30°\ V$, $\bar{V}_{X_C} = 60.14\angle -60°\ V$]

Q5. A circuit is consists of a resistance 6 Ω and a series capacitive reactance of 8 Ω.
A voltage e(t) = 141 sin(377t) is applied to the circuit.

(a) Find the complex impedance.
(b) Determine the rms and instantaneous values of current.
(c) Compute the power factor and power delivered to the circuit.
(d) Find the value of capacitance in farads.

[Ans.: (a) $Z = (6 - j8)\ \Omega = 10\angle -53.13°\ \Omega$, (b) $I_{rms} = 9.97\ A$, $i(t) = 14.1\sin(377t + 53.13°)\ A$, (c) $pf = 0.596$ leading, $S = 596.41 - j795.2$ VA, (d) $C = 331.56\ \mu F = 3.3156 \times 10^{-4}\ F$]

Q6. A Sinusoidal voltage is applied to three parallel branches yielding branch currents as follows:

$$i_1 = 14.14\sin(\omega t - 45^0),\ i_2 = 28.3\cos(\omega t - 60^0),\ i_3 = 7.07\sin(\omega t + 60^0)\ \text{A}$$

(a) Find the complete time expression for the source current.
(b) Draw the phasor diagram. Use the voltage as reference.

[Ans.: (a) $i(t) = 39.39\sin(\omega t + 15.10°)\ A$, (b) $\bar{I}_1 = 10\angle -45°\ A$, $\bar{I}_2 = 20\angle 30°\ A$, $\bar{I}_3 = 5\angle 60°\ A$, $\bar{I} = 27.85\angle 15.10°\ A$]

Q7. A voltage of $e(t) = 150\sin 1000t$ is applied across a series RLC circuit where

$$R = 40\,\Omega,\ L = 0.13\ H\ and\ C = 10\mu F.$$

(a) Compute the rms value of the steady-state current.
(b) Find the expression for the instantaneous voltage appearing across the inductor and capacitor terminals.
(c) Compare the rms value of the voltage appearing across L and C with that of the applied voltage and comment the result.
(d) Draw the complete phasor diagram showing all voltage and current.

[Ans.: (a) $\bar{I} = 2.12\angle -36.86°\ A$, (b) $v_C(t) = 300\sin(1000t - 126.86°)\ V$, $v_L(t) = 389.75\sin(1000t + 53.14°)\ V$, (d) $\bar{V} = 106\angle 0°\ V$, $\bar{V}_L = 275.6\angle 53.14°\ V$, $\bar{V}_C = 212.13\angle -126.86°\ V$]

Q8. In question, **Q7** determine the following.
(a) The real and reactive power supplied by the source.
(b) The reactive power supplied by the source.
(c) The reactive power of the capacitor.
(d) The reactive power of the inductor.
(e) The power factor of the circuit.

[Ans.: (a) $P_S = 179.79\ W$, $Q_S = 134.8\ VAr$ lagging, (b) $Q_S = 134.8\ VAr$ lagging, (c) $Q_C = 449.7156\ VAr$ leading, (d) $Q_L = 584.272\ VAr$ lagging, $pf = 0.8$ lagging]

Q9. In the network configuration shown in Fig. 3, find the current which flows through the $\overline{Z}_3$ branch. Use the nodal method.

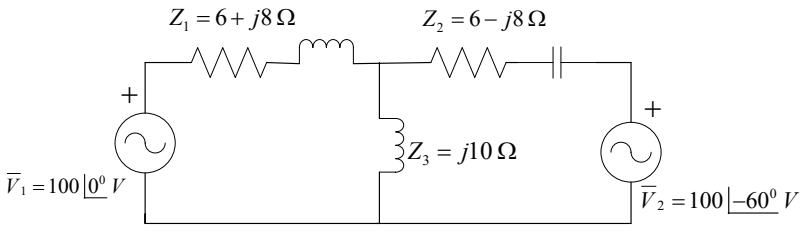


Fig. 3

[Ans.: $\bar{I} = 11.774\angle -80.2°\ A$]

Q10. A circuit has the configuration depicted in Fig. 4.

(a) Find the equivalent impedance appearing to the right of points a-b.
(b) Determine the value of the reactance X which makes the source current in phase with the source voltage.
(c) Should the reactance X of part (b) be inductive or capacitive? Find the required value of L and C.
(d) Compute the effective value of the source current for the condition described in part (b).

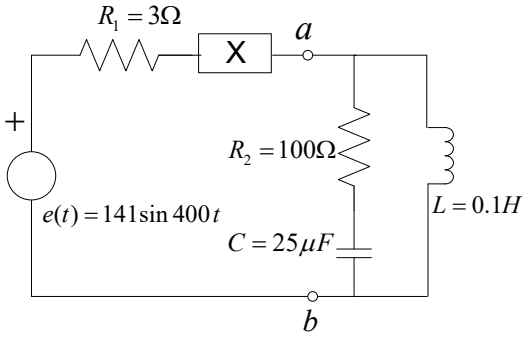


Fig. 4

[Ans.: (a) $Z_{ab} = (11.7647 + j47.05)\ \Omega$, (b) $X = -j47.05\ \Omega$, (c) X is capacitive, $C = 53.13\ \mu F$, (d) $i(t) = 9.5487\sin(400t)\ A$]

Q11. For the circuit shown in Fig. 5, calculate the following.

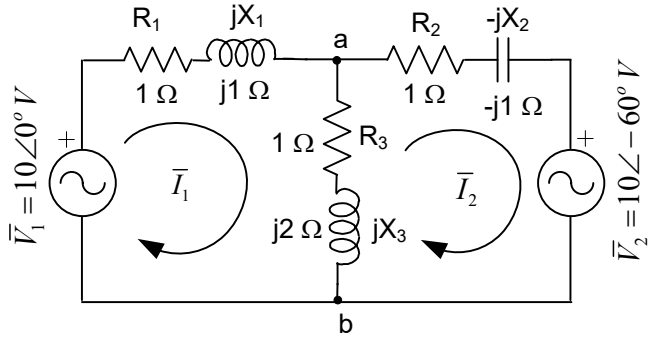


Fig. 5

(a) Using mesh method determine currents, $\bar{I}_1$, $\bar{I}_2$, and currents through $Z_3 = R_3 + jX_3$.
(b) Repeat (a) with nodal method.
(c) Find current through $Z_3 = R_3 + jX_3$ using Thevenin's equivalent circuit.

[Ans.: (a) $\bar{I}_1 = 1.58\angle 60°\ A$, $\bar{I}_2 = 6.05\angle 94.3°\ A$, $\bar{I}_{Z_3} = 4.82\angle -75°\ A$, (b) Same as (a), (c) $\bar{I}_{Z_3} = 4.82\angle -75°\ A$]

Q12. For the circuit shown in Fig. 5, calculate the following.

(a) Real, reactive and apparent powers and power factors of source 1 and source 2.
(b) What should be the value of $Z_3 = R_3 + jX_3$ for which the power transfer is maximum in Z_3. What is the value of the maximum power? For this maximum power what is the contribution of power from each source?

[Ans.: (a) $P_1 = 7.92\ W$, $Q_1 = 26\ VAr$, $S_1 = 60.5\ VA$, $pf_1 = 0.5$ leading, $P_2 = 54.57\ W$, $Q_2 = -13.73\ VAr$, $S_2 = 15.85\ VA$, $pf_2 = 0.9013$ lagging, (b) $Z_3 = 1\ \Omega$, $P_{max} = 46.65\ W$, $P_1 = 23.32\ W$, $P_2 = 23.32\ W$]